CUSTOMER CHURN ANALYSIS

Data Analytics Project | Workora Labs

1. Project Overview

This project analyzes a synthetic customer churn dataset to understand subscription cancellations, customer retention, revenue at risk and behavioral patterns.

2. Dataset Information

Metric	Value
Total Records	600
Total Columns	14
Churned Customers	247
Retained Customers	353

3. Key Business Metrics

Metric	Result
Churn Rate	41.17%
Retention Rate	58.83%
Mean-based CLV Proxy	INR 30,886.88
Median-based CLV Proxy	INR 24,687.57
Estimated Monthly Revenue at Risk	INR 240,102.48

CLV is estimated using average monthly charges multiplied by average tenure. It is a historical-value proxy, not a predictive future lifetime value. Revenue at risk is the sum of recorded monthly charges for churned customers.

4. Churn by Contract Type

Category	Total_Customers	Churned_Customers	Churn_Rate
Month-to-Month	316.0	189.0	59.81
One Year	164.0	33.0	20.12
Two Year	120.0	25.0	20.83

5. Churn by Tenure Group

Category	Total_Customers	Churned_Customers	Churn_Rate
0-12 Months	104.0	56.0	53.85
13-24 Months	110.0	50.0	45.45
25-48 Months	188.0	72.0	38.3
49-72 Months	198.0	69.0	34.85

6. Churn by Payment Method

Category	Total_Customers	Churned_Customers	Churn_Rate
Card	157.0	59.0	37.58
Net Banking	140.0	52.0	37.14
UPI	142.0	64.0	45.07
Wallet	161.0	72.0	44.72

7. Churn by AutoPay

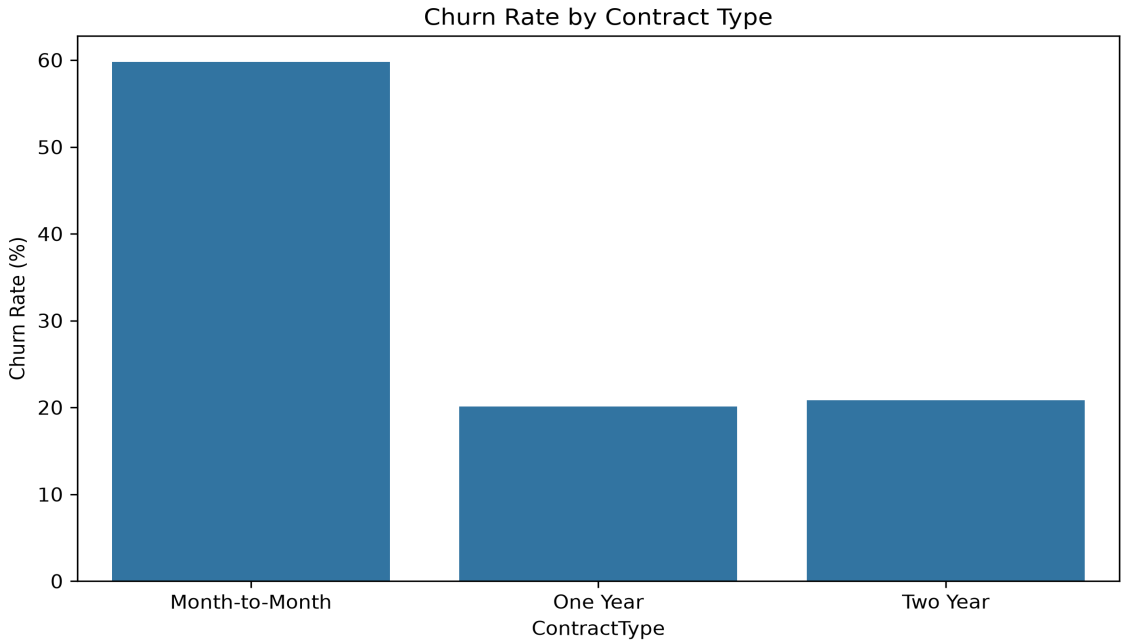
Category	Total_Customers	Churned_Customers	Churn_Rate
No	250.0	103.0	41.2
Yes	350.0	144.0	41.14

8. Behavioral Comparison

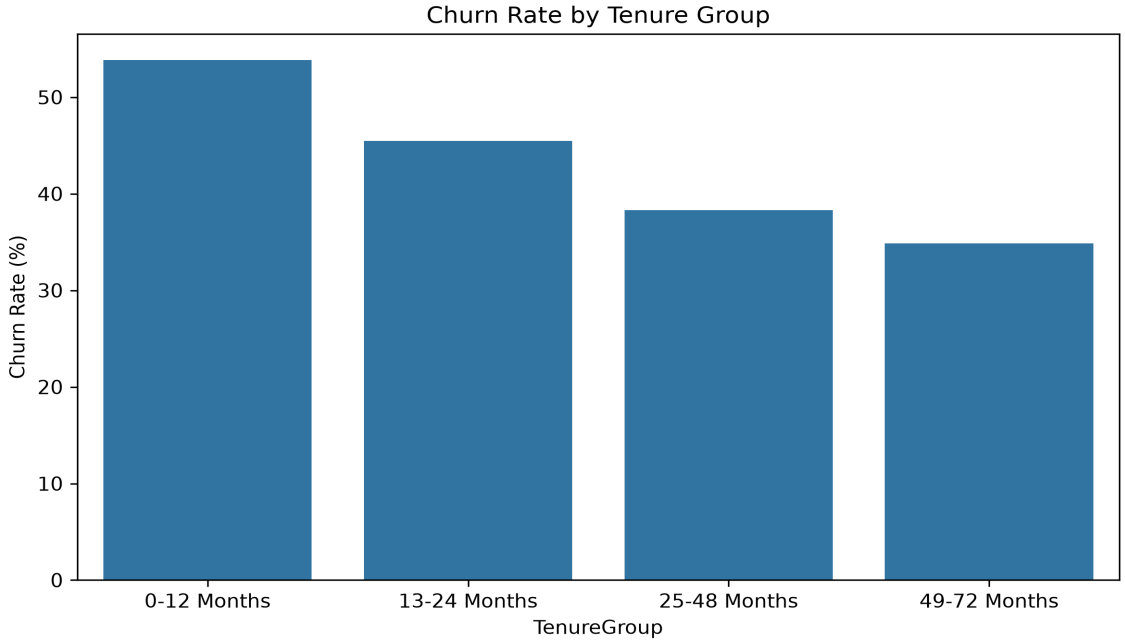
Category	Average_SupportTickets	Average_LatePayments
No	1.51	1.0
Yes	2.12	1.67

9. Data Visualizations

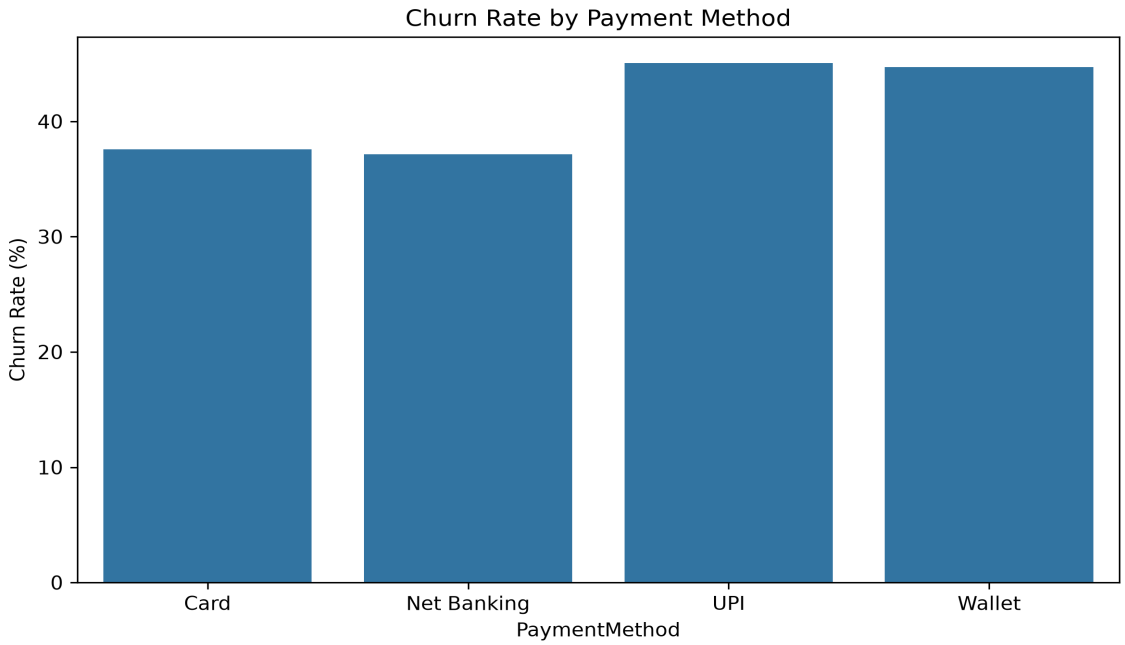
Churn Rate by Contract Type



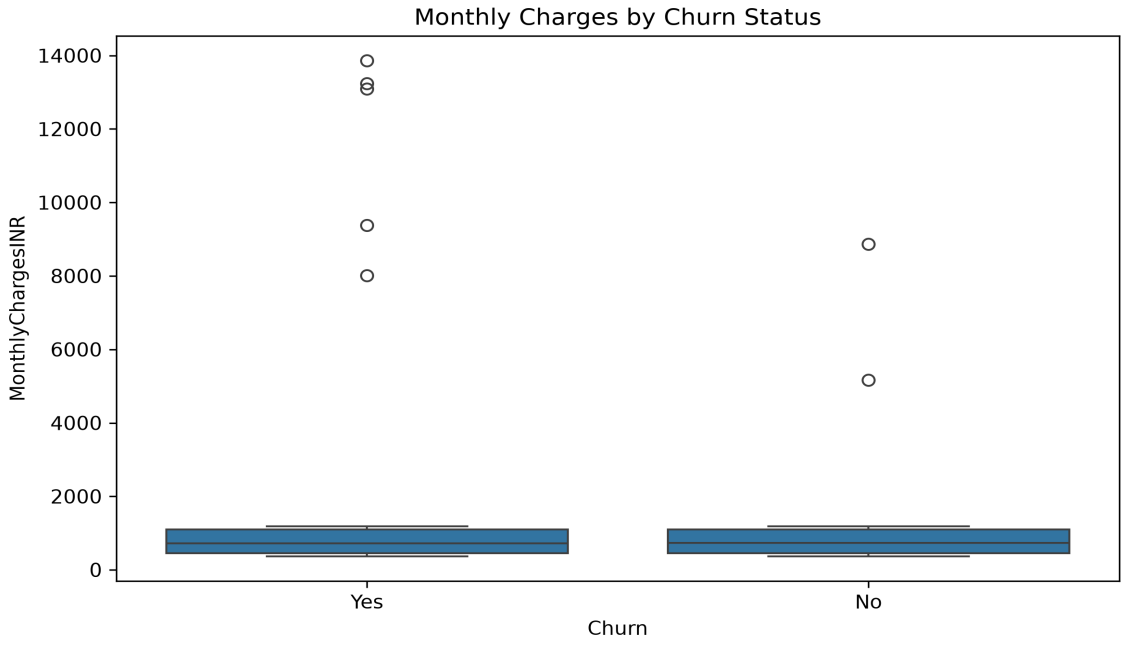
Churn Rate by Tenure Group



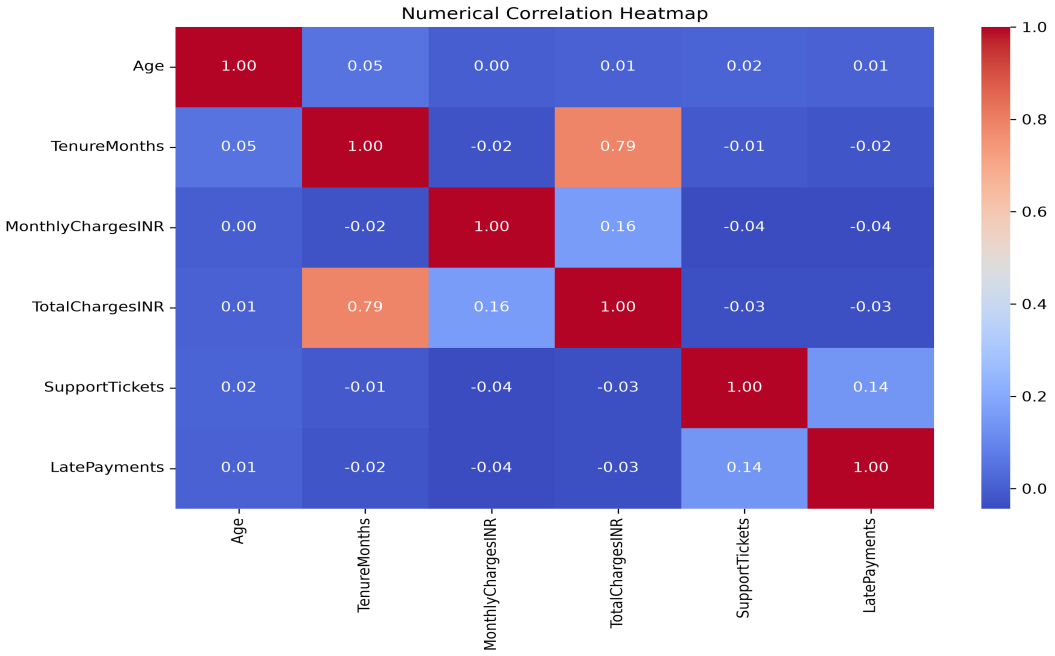
Churn Rate by Payment Method



Monthly Charges by Churn Status



Numerical Correlation Heatmap


Numerical Correlation Heatmap
Age
TenureMonths
MonthlyChargesINR
TotalChargesINR
SupportTickets
LatePayments
1.00 0.05 0.00 0.01 0.02 0.01
0.05 1.00 -0.02 0.79 -0.01 -0.02
0.00 -0.02 1.00 0.16 -0.04 -0.04
0.01 0.79 0.16 1.00 -0.03 -0.03
0.02 -0.01 -0.04 -0.03 1.00 0.14
0.01 -0.02 -0.04 -0.03 0.14 1.00
1.0
0.8
0.6
0.4
0.2
0.0

10. Key Findings

• Month-to-Month contracts show the highest churn rate at 59.81%.

• Customers with 0-12 months of tenure show the highest churn rate at 53.85%.

• UPI users show a churn rate of 45.07%, followed by Wallet users at 44.72%.

• Churned customers average 2.12 support tickets compared with 1.51 for retained customers.

• Churned customers average 1.67 late payments compared with 1.00 for retained customers.

• AutoPay users and non-users show almost identical churn rates.

11. Business Recommendations

• Introduce suitable retention offers for Month-to-Month customers.

• Strengthen onboarding and engagement during the first 12 months.

• Investigate repeated support complaints and improve resolution time.

• Use payment reminders and investigate late-payment patterns.

• Review UPI and Wallet transaction experiences.

• Monitor customer churn indicators regularly.

12. Limitations

The dataset is synthetic. The analysis identifies associations, not proven causes of churn. CLV is a historical proxy, and monthly revenue at risk is an estimate based on recorded charges.

13. Conclusion

The analysis identifies contract type, customer tenure, payment method, support tickets and late payments as useful areas for further customer-retention investigation.

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